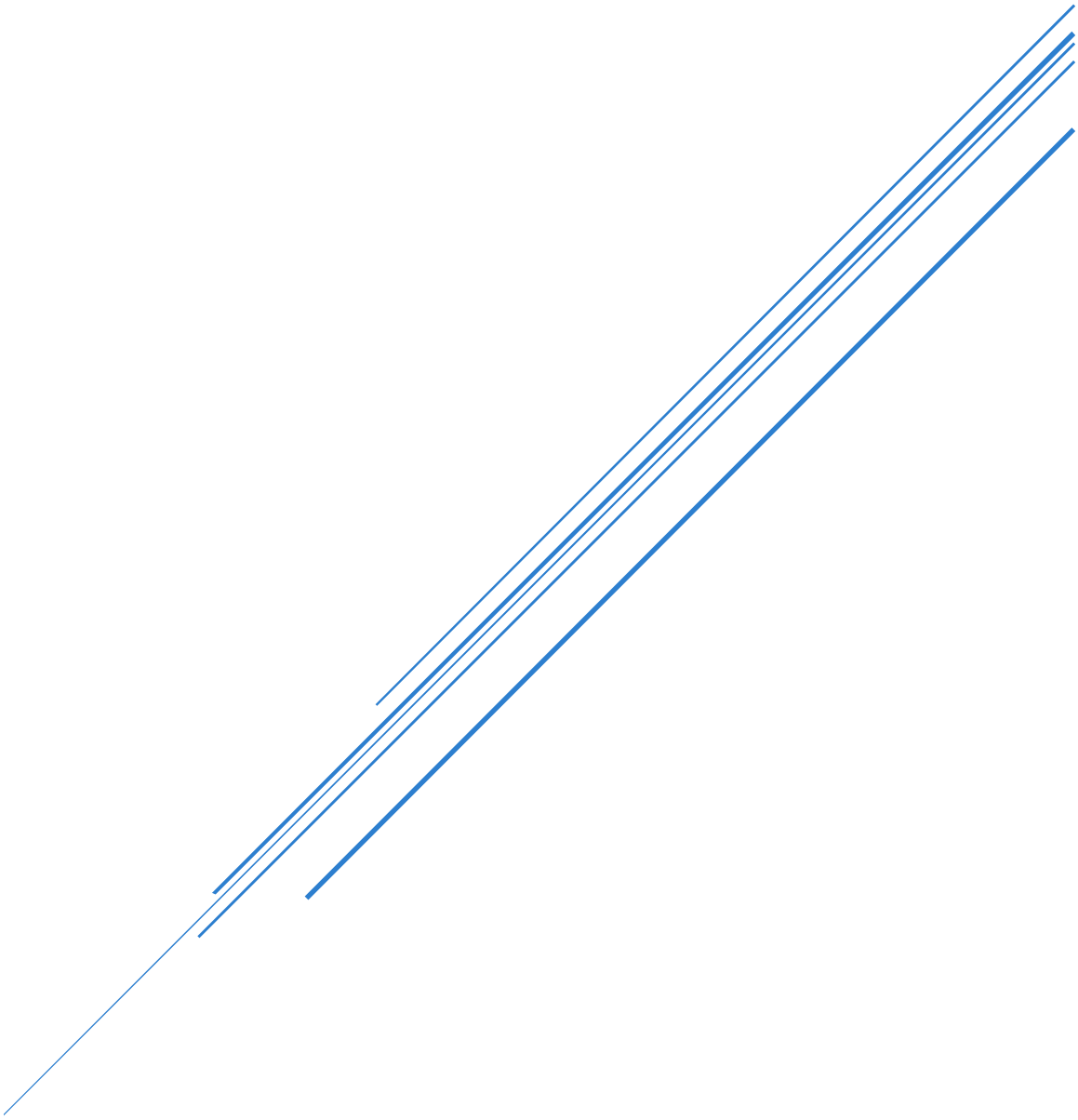
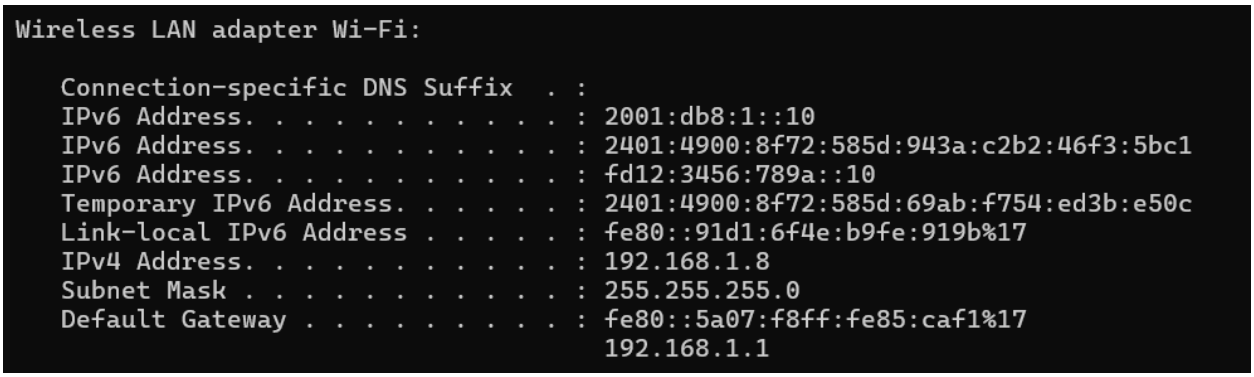
TROUBLESHOOTING TOOLS

Assignment - 6



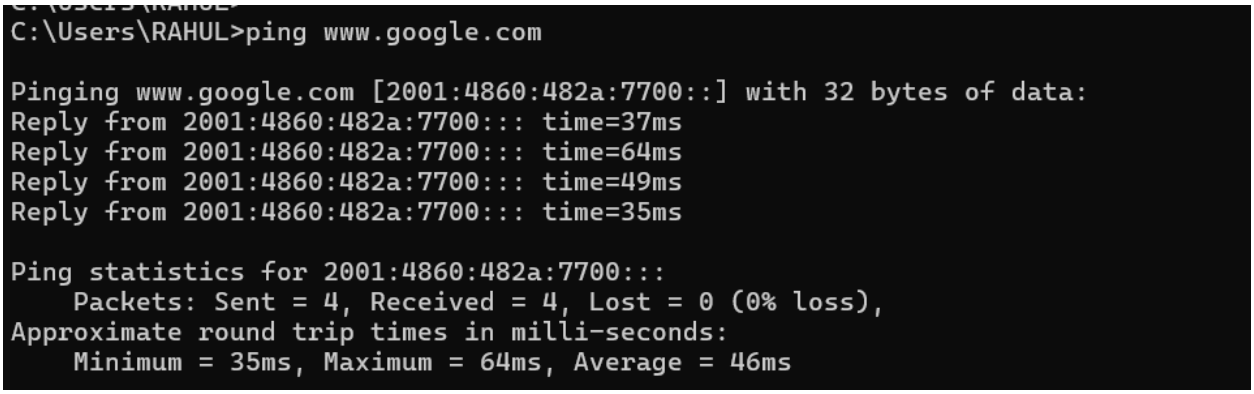
1. **Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.**

➢



2. **Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.**

➢



3. **Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.**

➢

```
C:\Users\RAHUL>tracert www.flipkart.com

Tracing route to e348826.dsca.akamaiedge.net [2.19.212.233]
over a maximum of 30 hops:

  1     3 ms     3 ms     2 ms  Unit [192.168.1.1]
  2    23 ms    20 ms    26 ms  10.240.170.222
  3     *        *        *     Request timed out.
  4    62 ms    22 ms    38 ms  125.21.143.117
  5   400 ms   439 ms    69 ms  116.119.109.250
  6  1048 ms   200 ms   278 ms  ae4.r03.border01.ccu01.fab.netarch.akamai.com [2.16.180.241]
  7   246 ms   268 ms   795 ms  lo1.r01.stem01.ccu01.fab.netarch.akamai.com [2.16.180.24]
  8   232 ms    82 ms   278 ms  lo1.r03.spine01.ccu01.fab.netarch.akamai.com [2.16.180.10]
  9   373 ms    83 ms   196 ms  lo1.r01.leaf01.ccu01.fab.netarch.akamai.com [2.16.180.4]
 10    63 ms    73 ms    96 ms  vlan100.r08.tor01.ccu01.fab.netarch.akamai.com [2.19.212.225]
 11   123 ms   101 ms    66 ms  a2-19-212-233.deploy.static.akamaitechnologies.com [2.19.212.233]

Trace complete.
```

4. **Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.**

➢

```
C:\Users\RAHUL>nslookup www.spotify.com
Server:  UnKnown
Address:  fe80::5a07:f8ff:fe85:caf1

Non-authoritative answer:
Name:    atc.spotify.map.fastly.net
Addresses:  2a04:4e42:59::810
          199.232.103.42
Aliases:  www.spotify.com
```

5. **A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check.**
   Hint: Think about checking both network connectivity and DNS resolution.

➢

No.	Command	Purpose
1	ipconfig	Checks the laptop's IP address, subnet mask, and default gateway.
2	ping zomato.com	Checks whether the laptop can reach Zomato's server and whether packets are responding.
3	nslookup zomato.com	Checks whether DNS can resolve Zomato's domain name into an IP address.